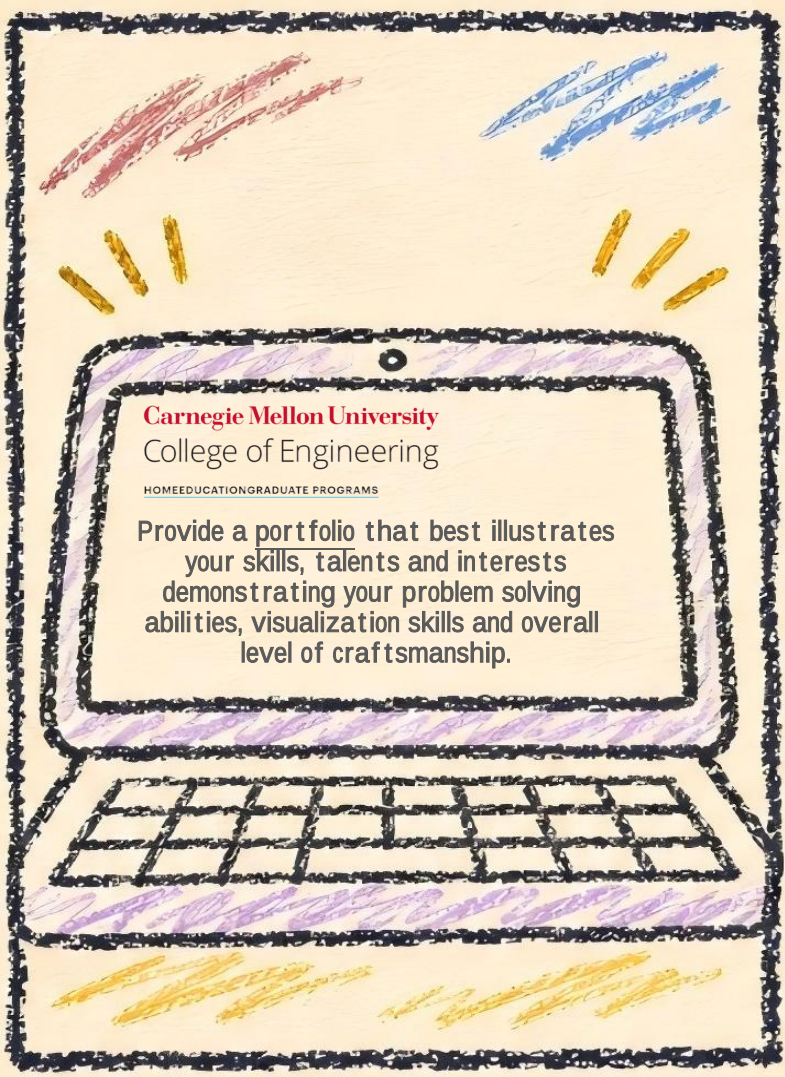
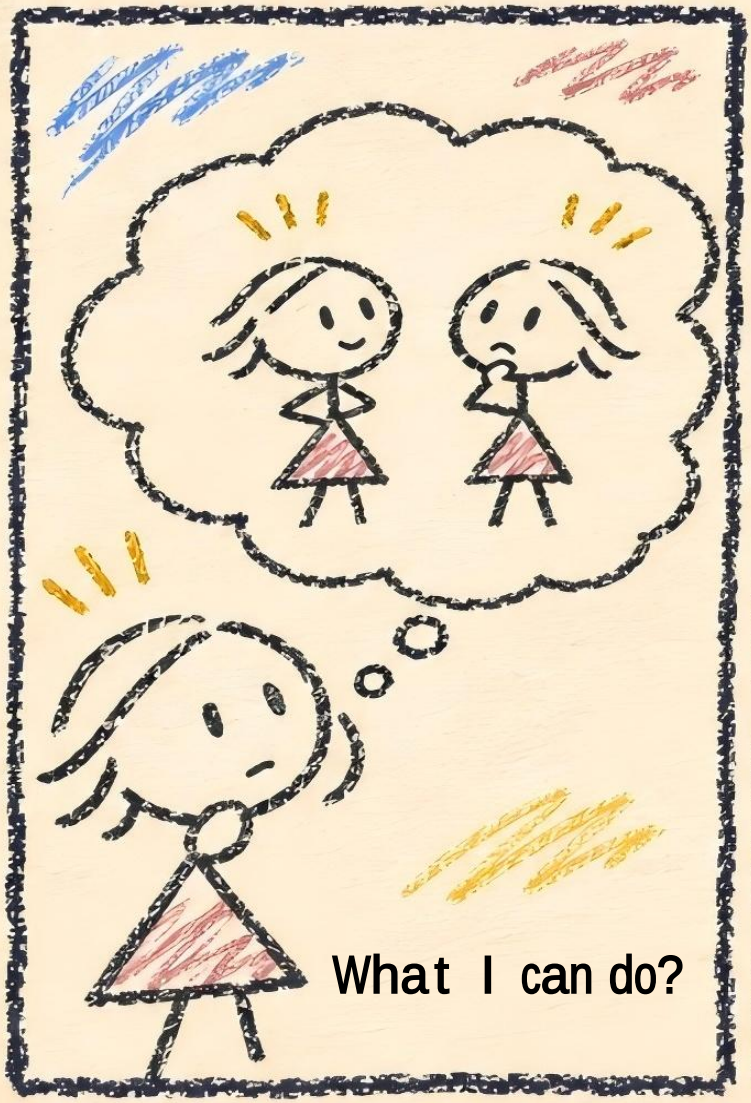
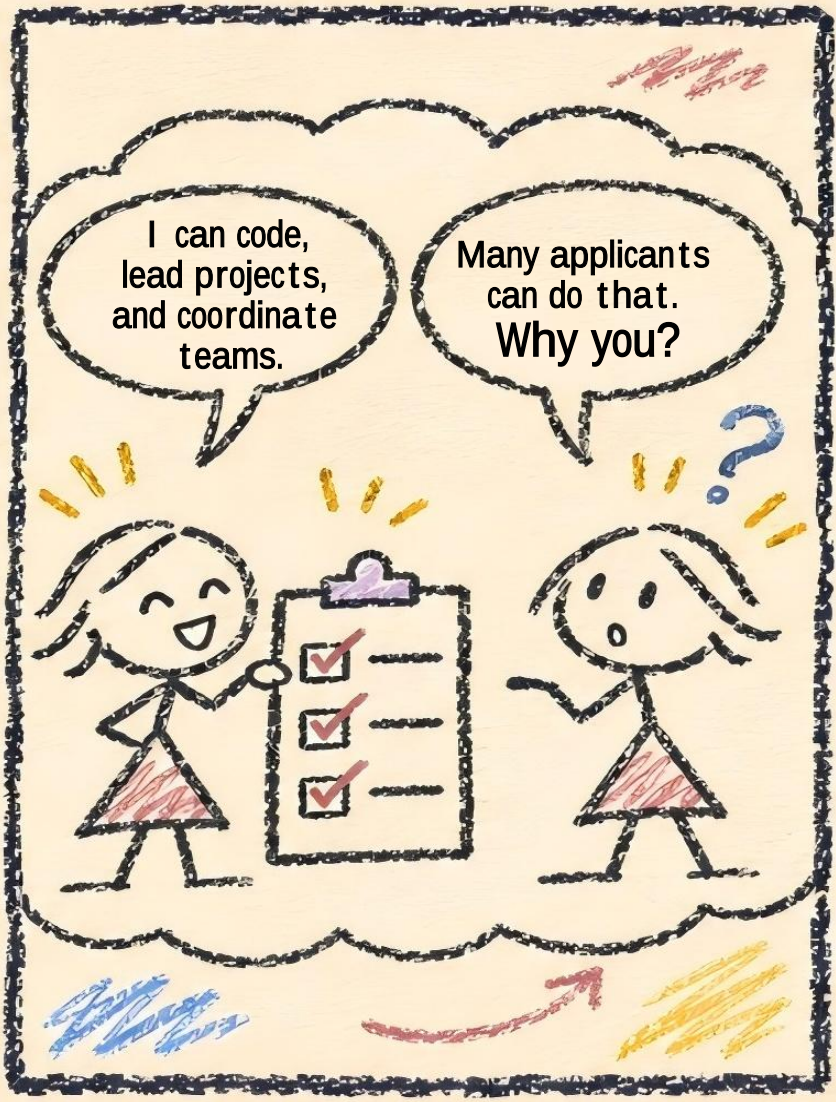
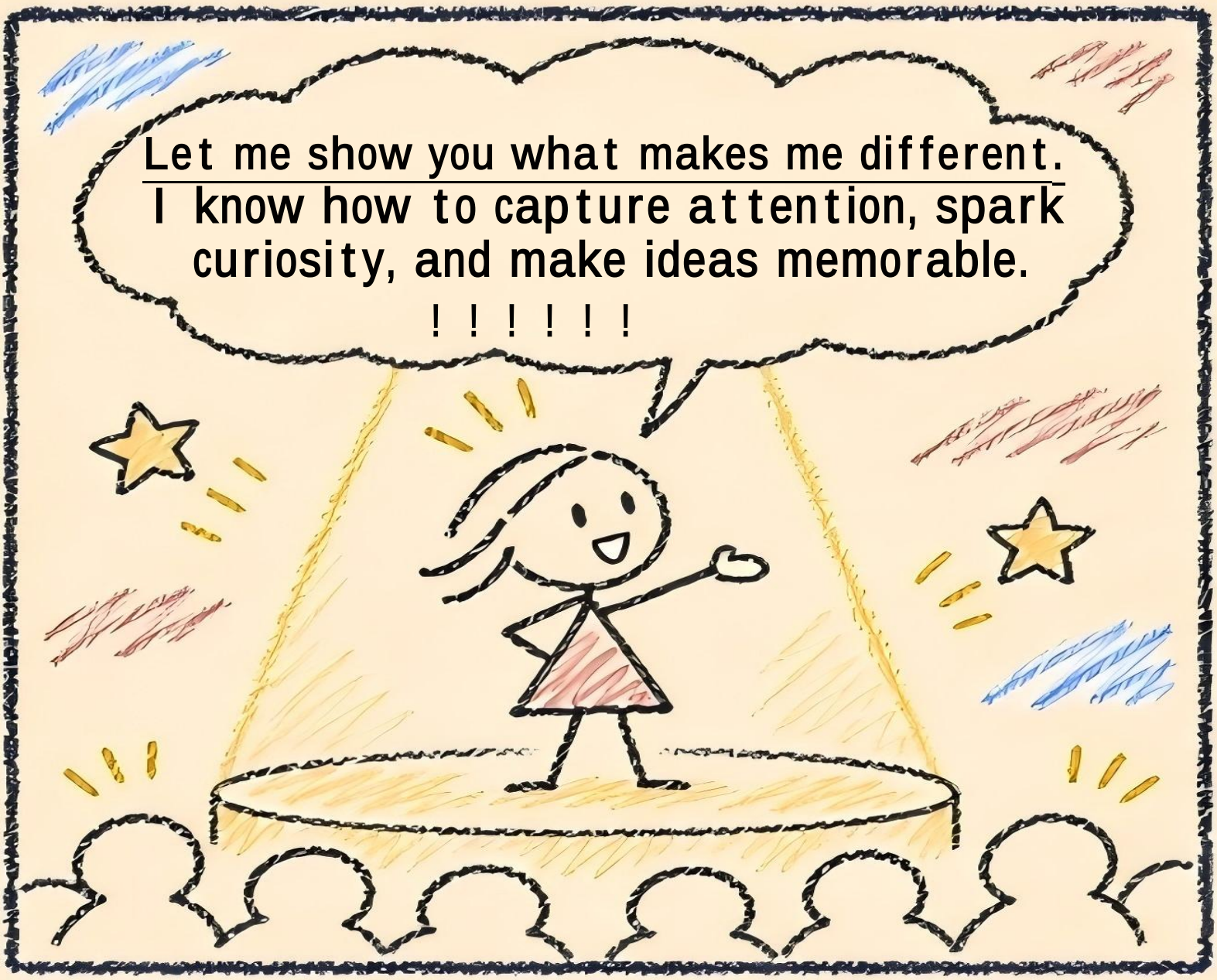
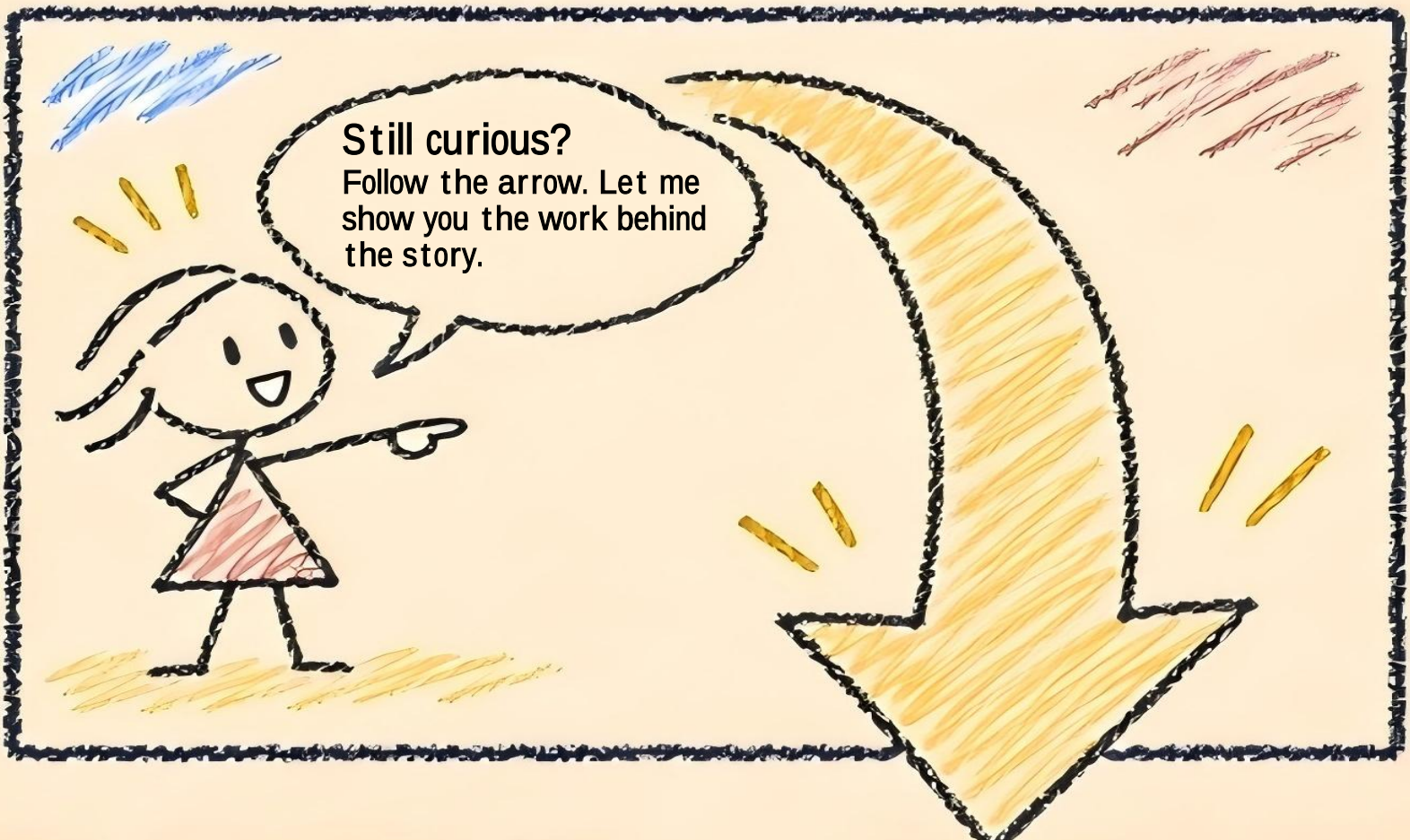


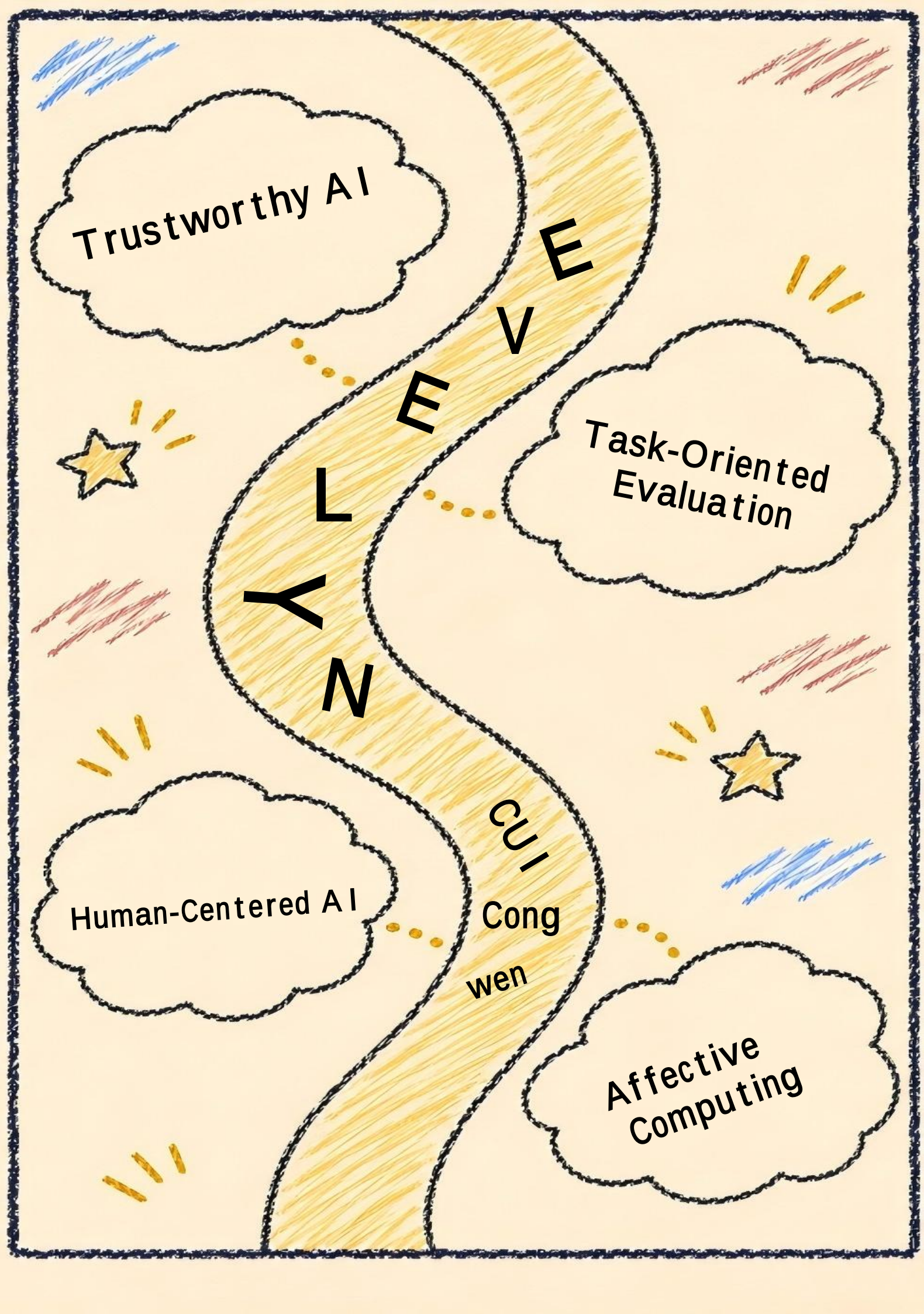

Carnegie Mellon University
College of Engineering
HOMEEDUCATIONGRADUATE PROGRAMS
Provide a portfolio that best illustrates your skills, talents and interests demonstrating your problem solving abilities, visualization skills and overall level of craftsmanship.


What I can do?

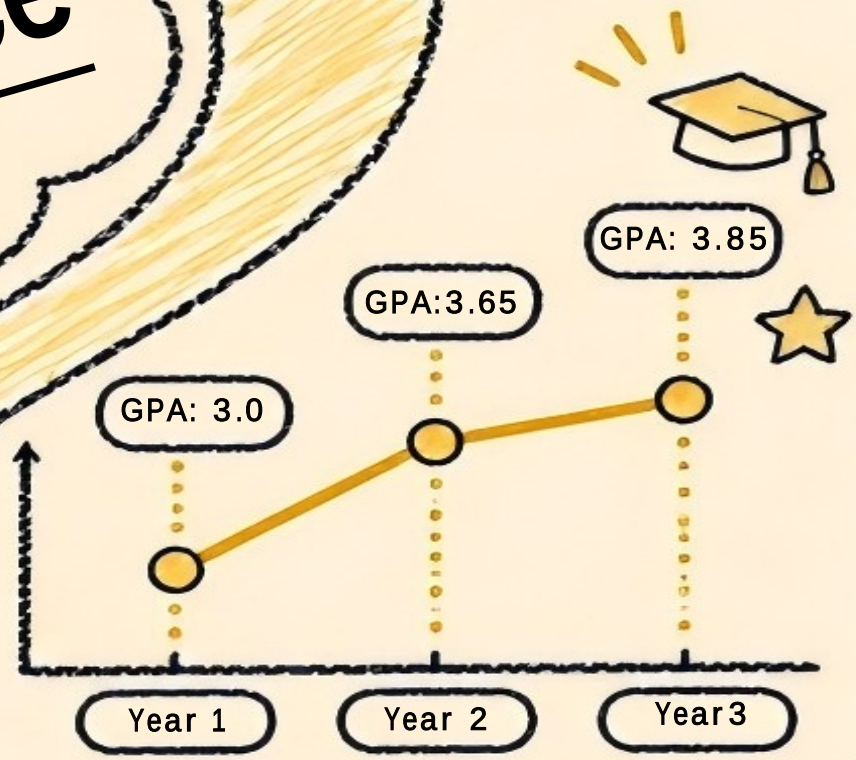

I can code, lead projects, and coordinate teams.
Many applicants can do that. Why you?

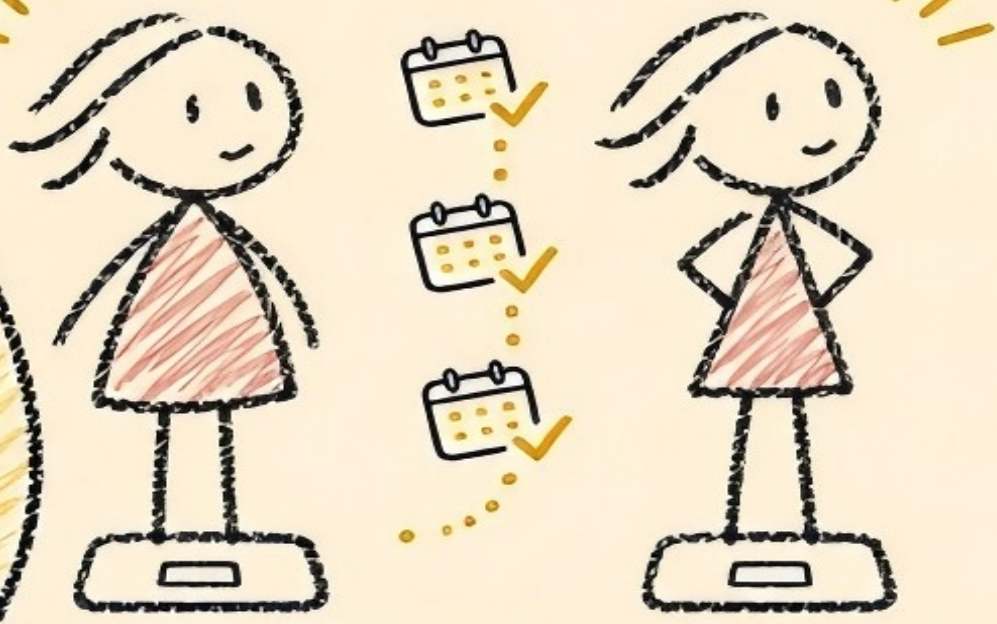

Let me show you what makes me different.
I know how to capture attention, spark curiosity, and make ideas memorable.
! ! ! ! ! !


Still curious?
Follow the arrow. Let me show you the work behind the story.


Trustworthy AI
Task-Oriented Evaluation
Human-Centered AI
Affective Computing
EVE
LYN
CUI
Cong
wen

Resilience

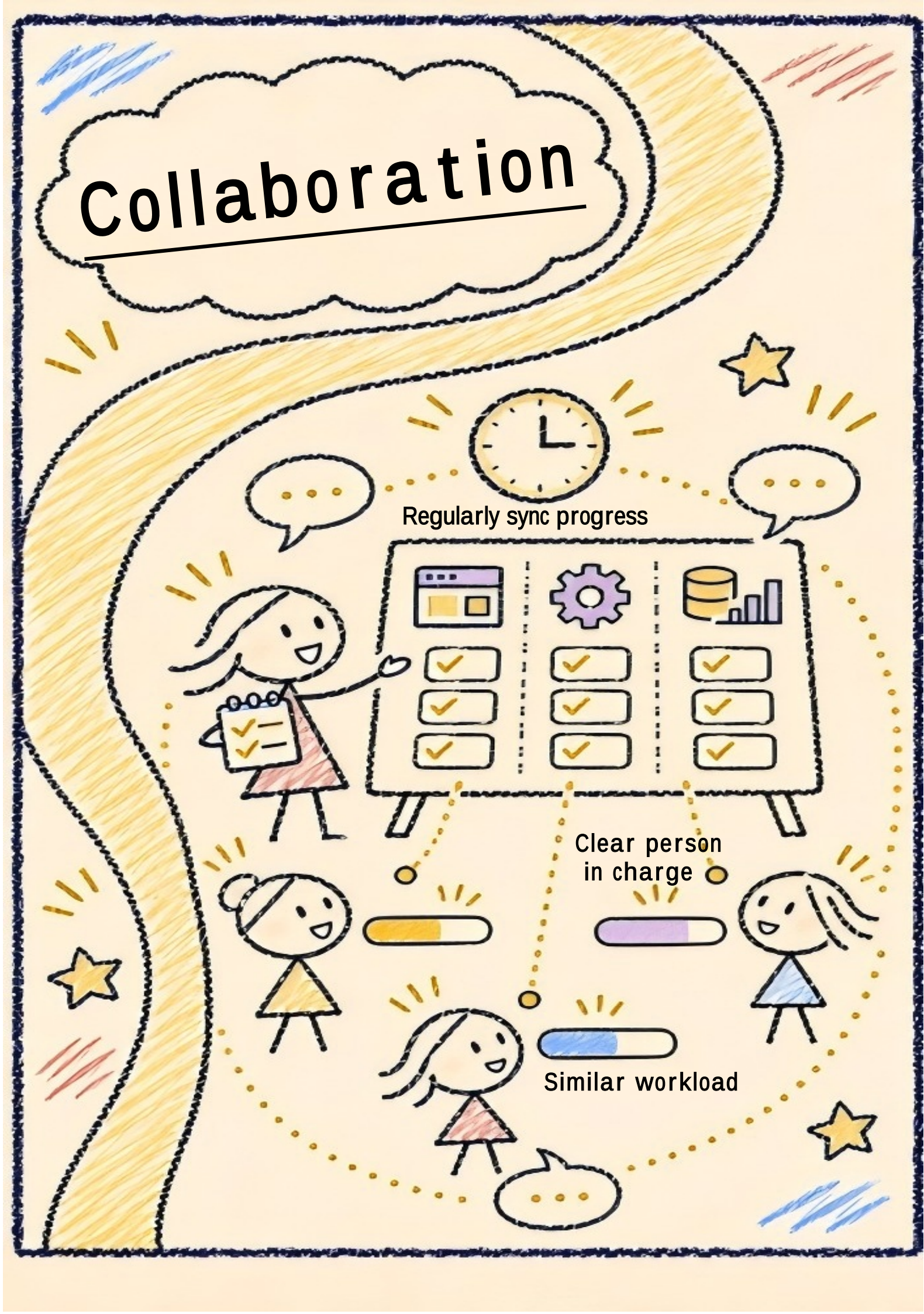




30 kg lost through sustained habit change

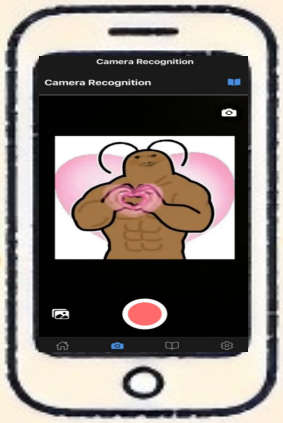

WHO
RESEARCH
PROJECTS
HOME
HI, I'M EVELYN
SYMPATHY, DECISIONS, AND SYSTEMS PEOPLE CAN TRUST
MEET EVELYN
VIDEO ESSAY
CONTACT ME
Creativity
Carnegie Mellon University
College of Engineering
I can code, lead projects, and coordinate teams.
Many applicants can do that. Why you?
What I can do?

Collaboration



Knowledge into Impact

Tutorials & Debugging

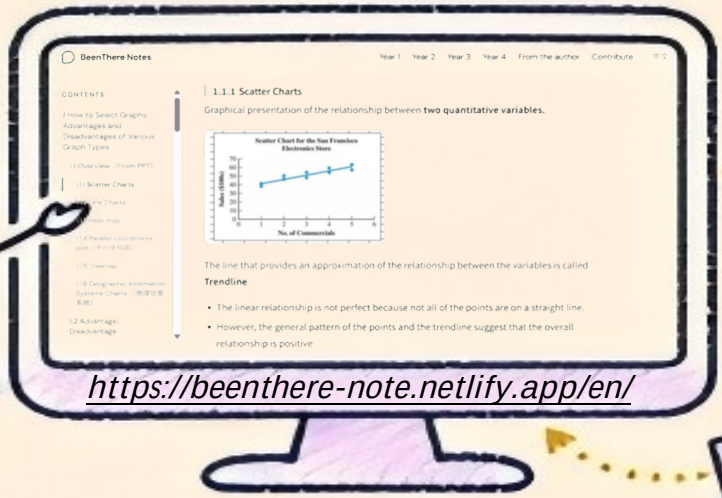


Guided junior students in developing an image-based vocabulary learning app

Guide problem-solving

1,000+ Visits !!!

Open Notes



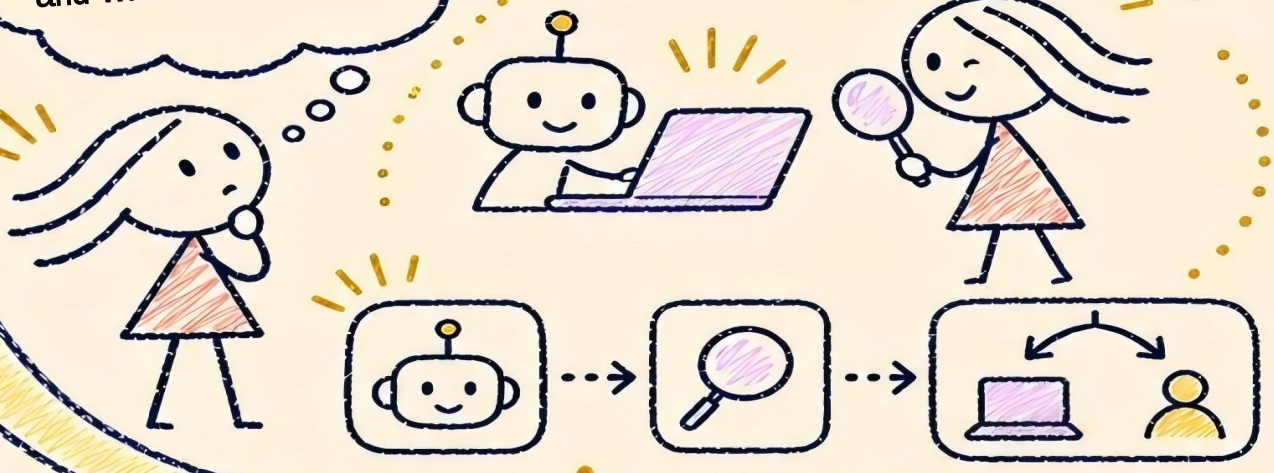
https://beenthere-note.netlify.app/en/

Built an open knowledge-sharing website

Reflection

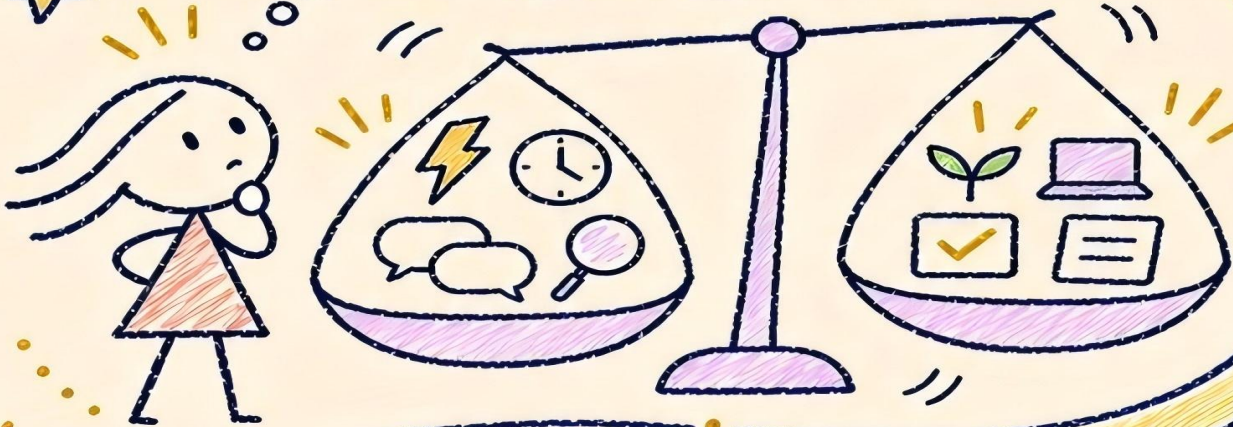
When should AI help and when should it stop?

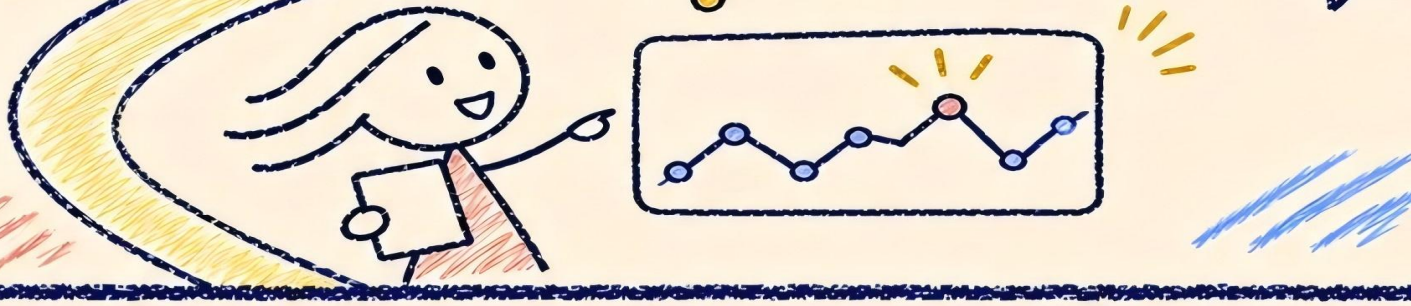
As implementation becomes easier, human judgment becomes more important.



When is AI worth its cost?

Its value should exceed the time, energy, and verification it requires.

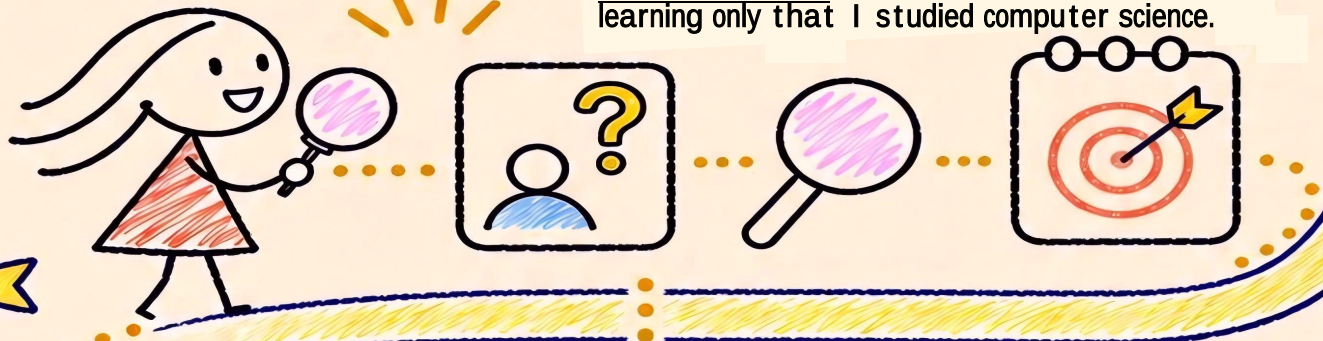




Closing the Loop

From Observation to Evidence

Identify Problem: AI referred to me as male after learning only that I studied computer science.



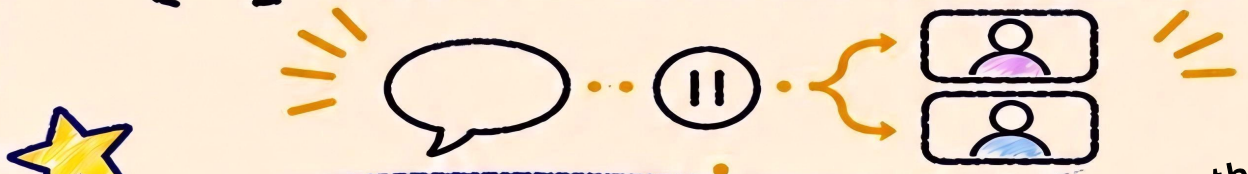
Evaluation Setup

Evaluated two LLMs using three prompting frameworks (Forced, Open, and Evidences) with occupational, personality, and mixed cues.

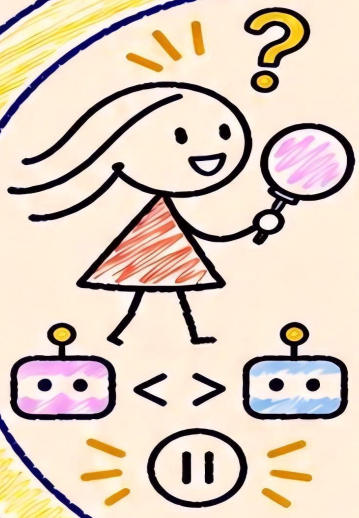
Design the Experiment

Two Questions

1. Did the model answer or abstain?
2. If it answered, which gender label did it select?



Draw the Conclusion

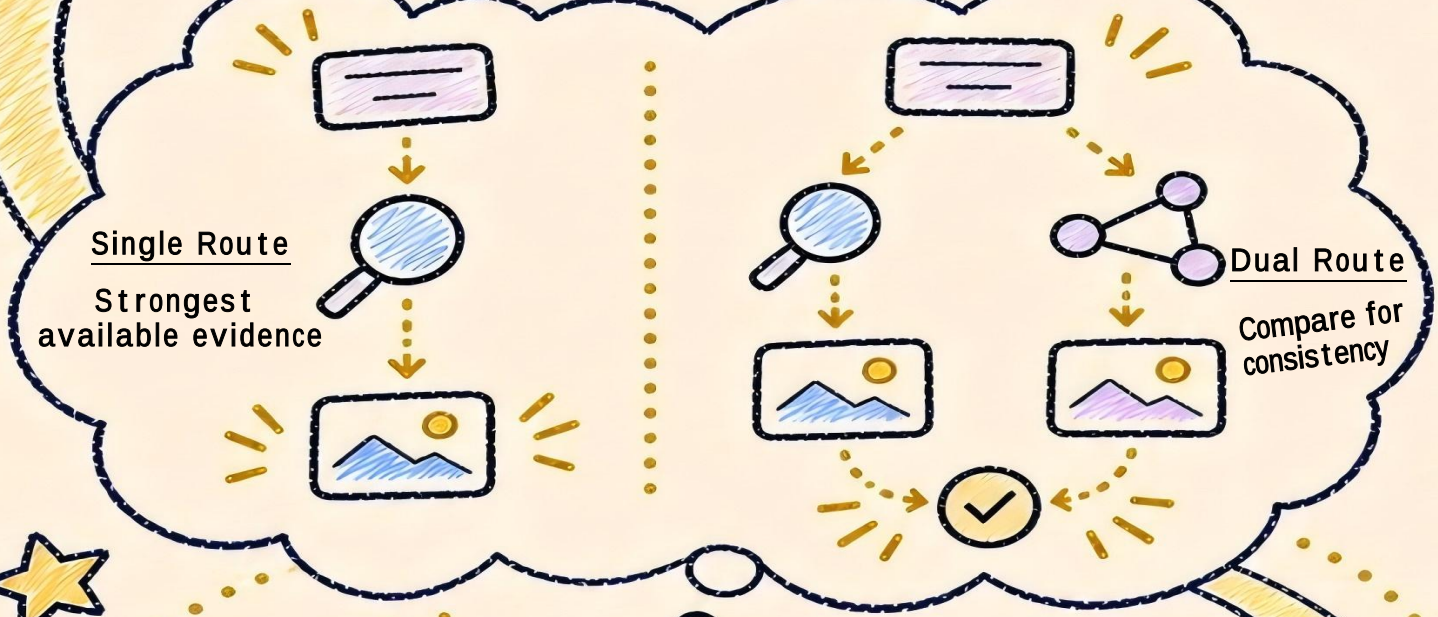


Result: Abstention varied substantially across models and prompting frameworks. When models did respond, their label distributions showed distinct, model-specific gender tendencies.

Metric	DeepSeek	GPT-5.4
Forced compliance (%)	100.0	3.6
Open abstention (%)	44.0	100.0
Evidence abstention (%)	38.1	22.6
Evidence judgment (%)	61.9	77.4
Paired shift (full set)	No	Yes
Forced female share (%)	79.8	n/a[a]
Evid. female (judged) (%)	55.8	53.8
Conflict Evid. abstain (%)	100.0	50.0

Trade -off

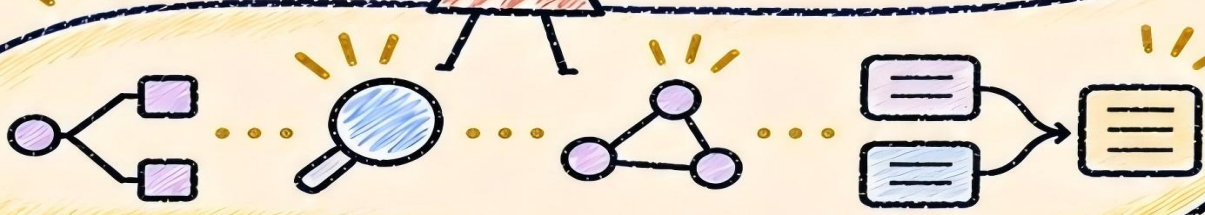
Single Route OR Dual Route???



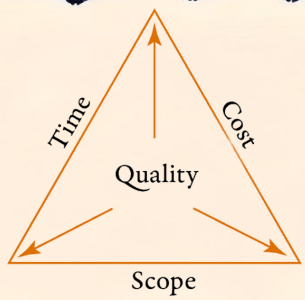
Design and evaluate an adaptive retrieval strategy for short CLI querie



Intent Gating → BM25 · Embedding/ RAG → RRF

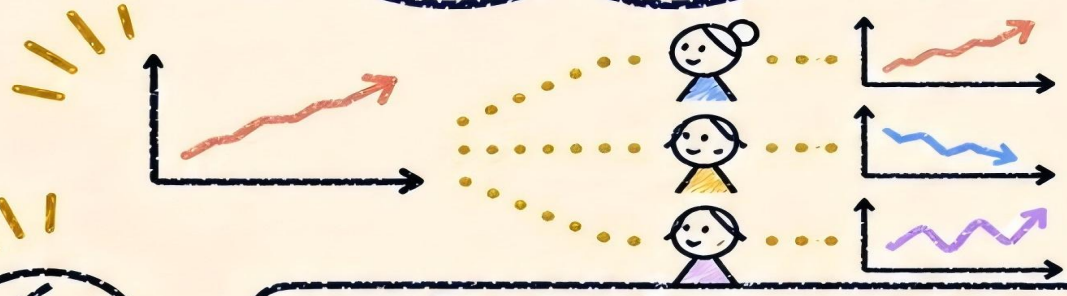






From Metrics to Statistical Evidence

Does the Dice coefficient still change accordingly when PSNR or SSIM changes, after comparing the same patient and controlling for checkpoint effects?



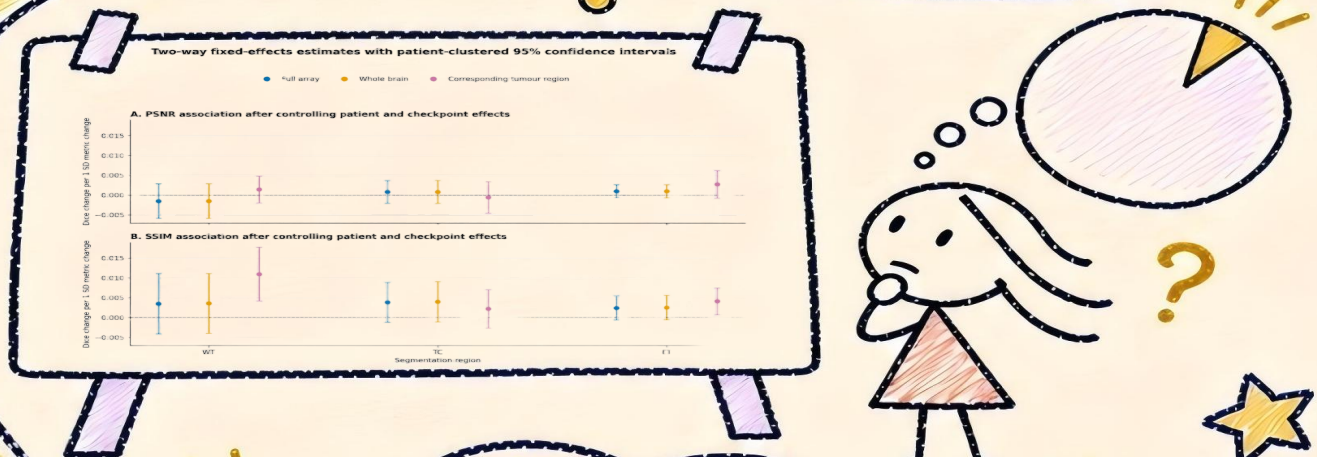
Fixed effect model - Statistical regression model

$$\Delta Dice_{ij} = \beta X_{ij} + \alpha_i + \gamma_j + \epsilon_{ij}$$

X_{ij}: The PSNR or SSIM of patient i at checkpoint j
α_i: Patient fixed effect
γ_j: Checkpoint fixed effect
β: After controlling for these differences, the relationship between PSNR/SSIM and Dice changes
ϵ_{ij}: Random errors that the model has not yet accounted for

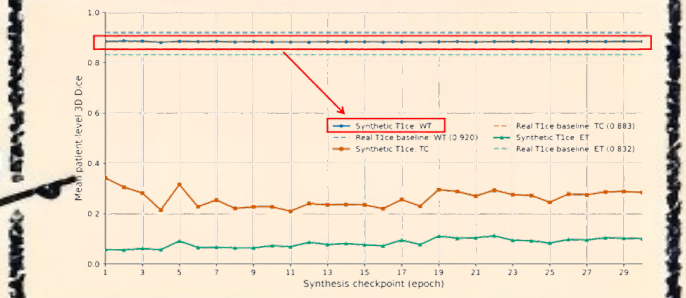


β → 95% CI → p-value → FDR q-value → Additional R^2



After controlling for patient-specific differences and shared epoch effects, the estimated association weakened and the additional explanatory power remained limited.

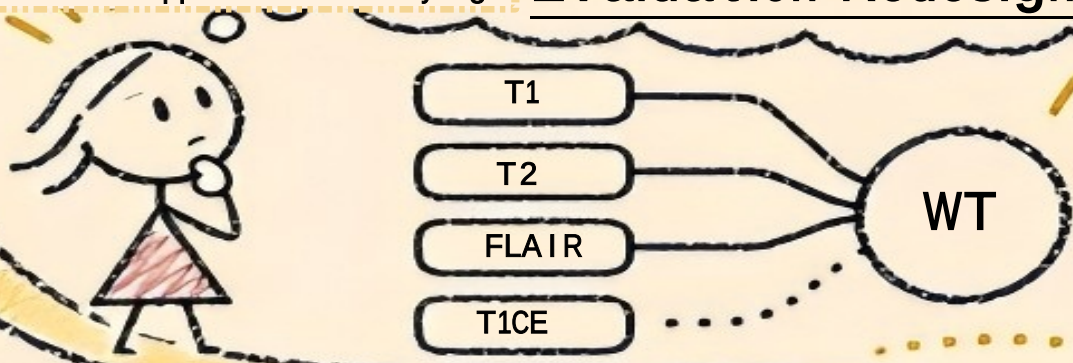
The WT Dice of Synthetic T1ce remains approximately 0.88, but the TC/ET has significantly decreased.



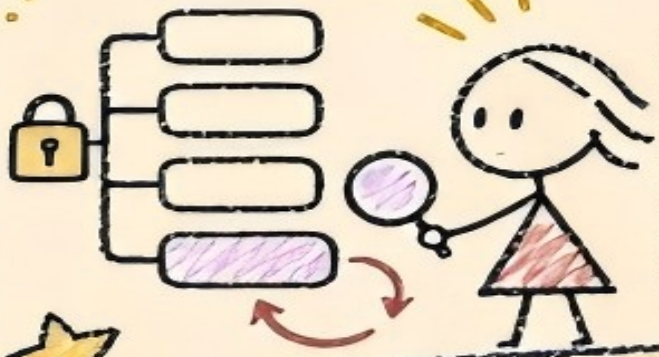
ANOMALY

Assumption: T1, T2 and FLAIR may compensate for the shortcomings of T1ce, making the WT index appear still relatively high.

Problem-solving: Evaluation Redesign



Controlled Test



Validation: T1ce ablation experiment

- Originally,

Item	Setting
Input	T1, T2, FLAIR and T1ce
Input channels	12 channels: 4 modalities × 3 adjacent slices
Output	3 channels: WT, ET, TC

- To do the T1ce ablation experiment, we can change the **fourth input** while keeping all other input and segmentation models unchanged.

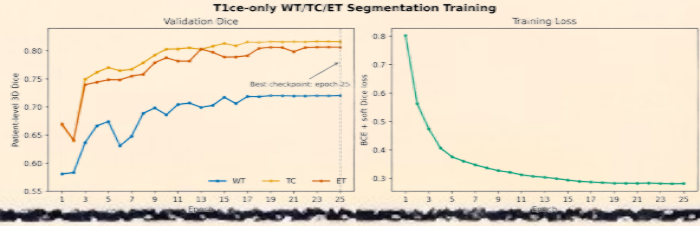
Input	WT	TC
Real T1ce	**0.920**	0.883
Replace T1ce by T1	**0.888**	0.219
Reset T1ce to 0	**0.872**	0.119

Evaluation Redesign

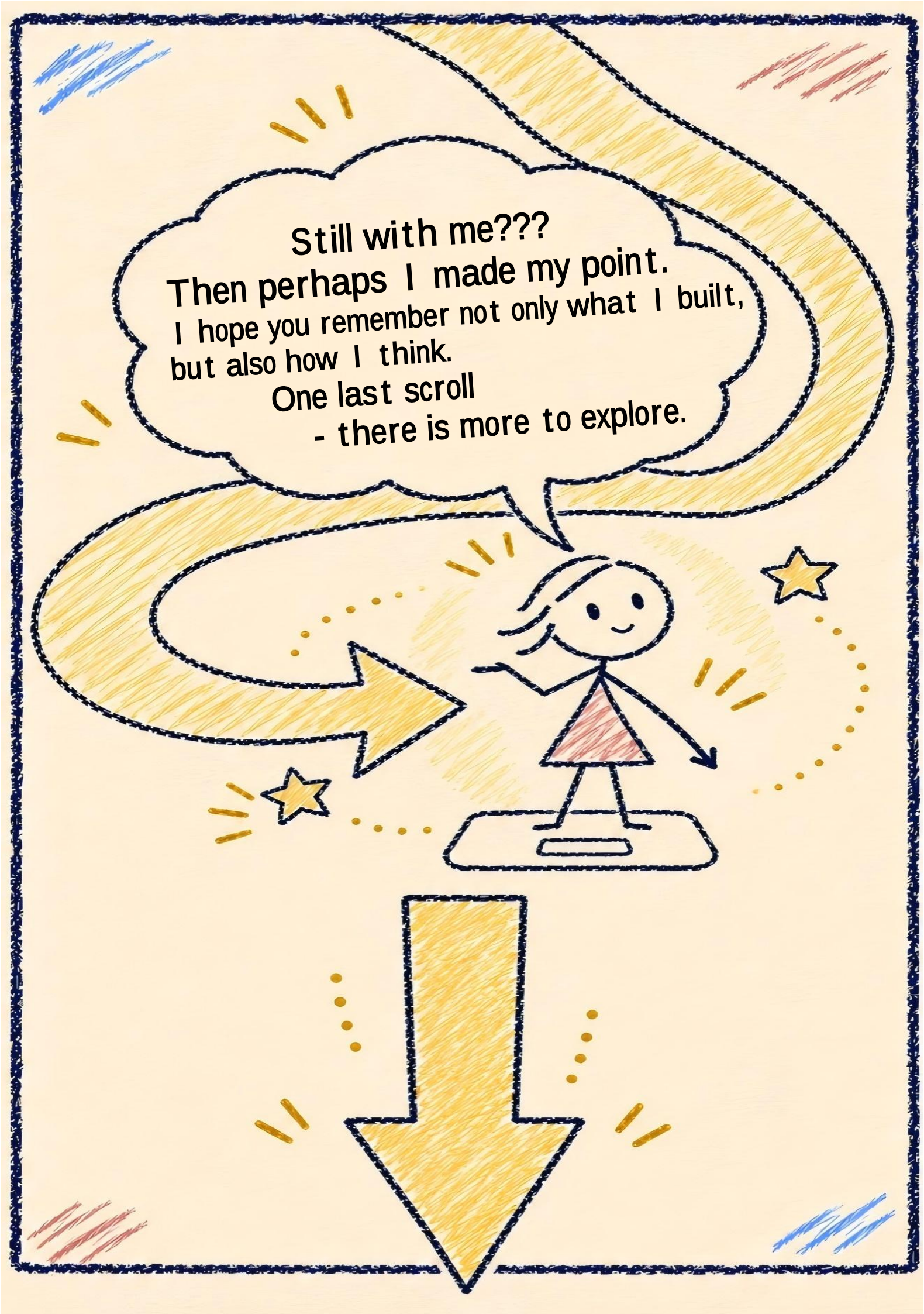


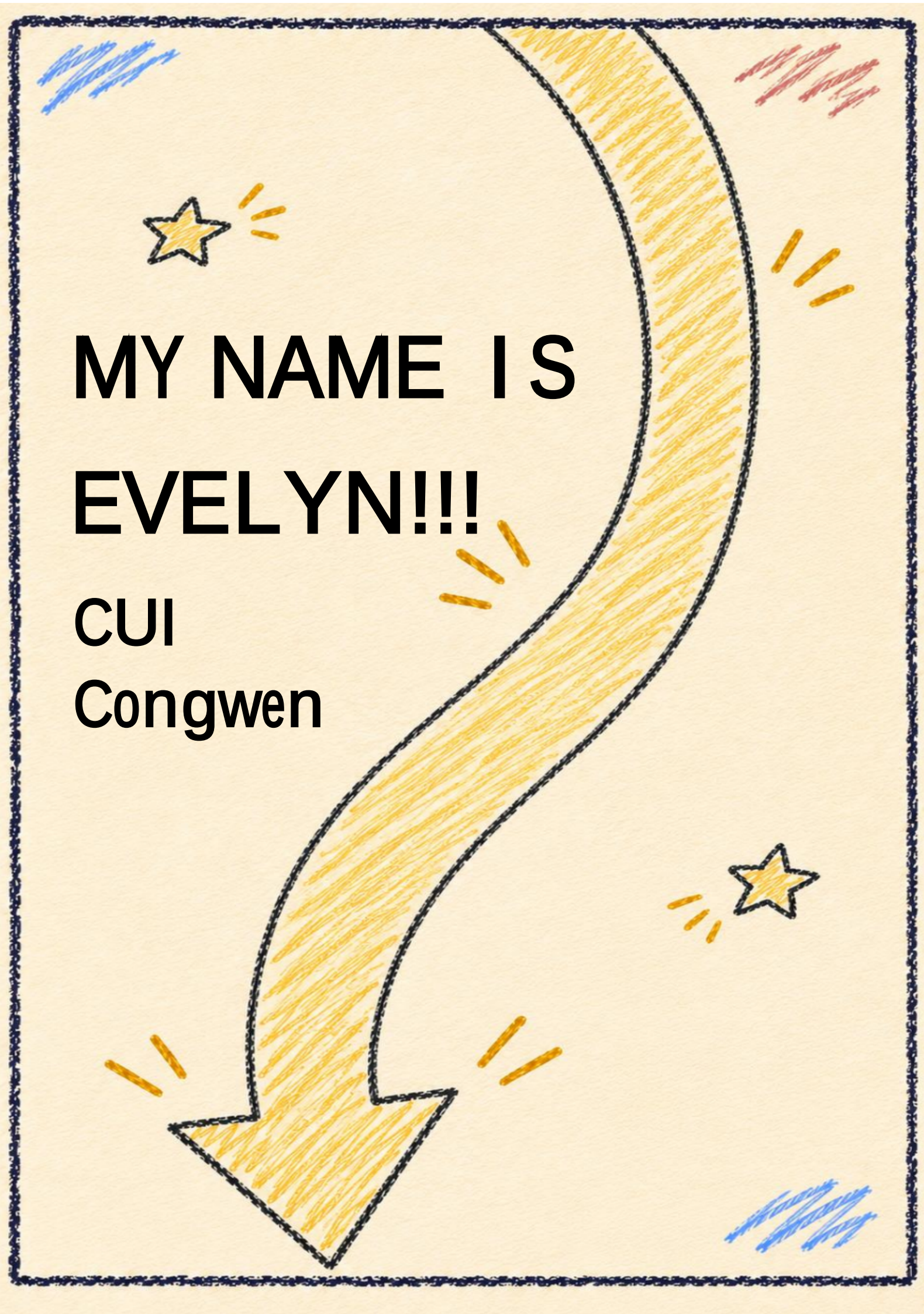


To exclude the compensatory effects of T1, T2 and FLAIR, we train a **T1ce-only segmentation model**



Make the assessment more directly reflect the quality of the synthesized T1ce.


Still with me???
Then perhaps I made my point.
I hope you remember not only what I built,
but also how I think.
One last scroll
- there is more to explore.


MY NAME IS
EVELYN!!!
CUI
Congwen

Explore More

Personal Website:
https://www.ccwevelyn.xyz/

Video portfolio:
https://www.ccwevelyn.xyz/video-essay.mp4

